

SANIKA TARE

AI SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Results-driven Software Engineer and Computer Engineering student (B.E., 2027) who builds sustainable, efficient web applications with exceptional user interfaces. Strong in full-stack development, system design, and data-driven problem solving, with measurable results across an industry internship and two benchmarked projects. Collaborative leader who takes ownership and delivers reliable, user-focused solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL

AI/ML: Machine Learning, Deep Learning, NLP, Transformers, XGBoost, Predictive Analytics, Feature Engineering, SHAP, Prompt Engineering

GenAI & RAG: LLMs, RAG, AI Agents, Multi-Agent Systems, LangChain, FAISS, BM25, ChromaDB, Embeddings, Hybrid Search, Reranking, Gemini

Backend: FastAPI, Node.js, Express.js, REST APIs, Microservices, JWT, RBAC, SQLAlchemy, Pydantic

Frontend & UI: React.js, React 19, TypeScript, HTML5, CSS3, Tailwind CSS, Vite, Recharts, Responsive Design

Databases & Cloud: PostgreSQL, MongoDB, pgvector, AWS, Docker, Docker Compose, Nginx

CS Fundamentals: DSA, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems, System Design

Leadership & Soft Skills: Team Leadership, Collaboration, Technical Communication, Ownership, Problem Solving

EXPERIENCE

Tata Technologies— AI/ML Engineering Intern, Vehicle IQ Digital Twin | Pune, India

June 2026 – August 2026

Python | XGBoost | SHAP | FastAPI | React 19 | LangChain | ChromaDB | Gemini | RAG | Docker | Nginx

- Developed an AI Vehicle Digital Twin for health monitoring, predictive maintenance, OBD-II diagnostics, and trip intelligence.
- Unified 8 automotive data sources into a 70,000-row, 209-feature dataset for health analysis and failure prediction.
- Built an XGBoost failure classifier (0.998 ROC-AUC, 99.4% recall) and Remaining Useful Life (RUL) estimation.
- Applied SHAP to identify torque, rotational speed, and tool wear as key failure drivers.
- Designed 8 independent FastAPI services with a React 19 dashboard, containerized with Docker Compose and Nginx.
- Implemented a LangChain, ChromaDB, and Gemini RAG pipeline for document-grounded diagnostic guidance.
- Delivered trip intelligence combining vehicle health, route, weather, and fuel data into travel advisories.

PROJECTS

HomeIQ — Multi-Agent AI Home Intelligence Platform

Python | FastAPI | PostgreSQL | pgvector | LangChain | Gemini | BioBERT | RAG | Docker

- Built a multi-agent platform with 8 domain agents (Kitchen, Laundry, Maintenance, Finance, Vehicles, Documents, Health, Travel) and tool-based workflows.
- Delivered document intelligence: 100% classification and extraction F1 on a 12-document benchmark.
- Designed grounded RAG with 100% citation correctness and 0% hallucination across 8 domain queries.
- Developed biomedical NLP reaching 99.7% biomarker F1, 98.9% PubMedQA accuracy, and 99.4% interaction F1.
- Verified 100% agent routing, 100% Human-in-the-Loop policy enforcement, zero cross-tenant leakage, 20/20 tables.

VedaWise — Explainable RAG & Knowledge Retrieval System

Python | FAISS | BM25 | RRF | Transformers | Reranking | RAG | LLMs

- Architected an explainable RAG system over 10,546 verses using BM25, 384-dim embeddings, FAISS, RRF, reranking, and evidence filtering.
- Improved retrieval on a 92-question benchmark to 90% Recall@10, 0.7838 MRR, and 0.7120 nDCG@5.
- Implemented citation verification and abstention: 100% citation precision and abstention accuracy.
- Reached 93.75% answer correctness at 115.7 ms average query latency while enforcing evidence boundaries.

ACHIEVEMENTS

- Filed a patent for an ML-based predictive Li-Fi/Wi-Fi handover system using smartphone motion sensors.
- Presented AI/ML predictive-systems research at KSHITIJ 2026.

EDUCATION

Pimpri Chinchwad College of Engineering | Pune, Maharashtra

2023 – 2027 (Expected)

Bachelor of Engineering in Computer Engineering | Coursework: AI, ML, DSA, DBMS, Networks, Distributed Systems

CERTIFICATIONS

- Databricks — Generative AI Fundamentals
- AWS — Cloud Practitioner Essentials
- The AI Engineer Course — Bootcamp
- AICTE — Generative AI Virtual Internship
- Data Structures using C and C++
- IIT Guwahati — Summer Analytics